User Manual for "The Journey of the Eagle"

1. Introduction

Welcome to The Journey of the Eagle, a social augmented reality (AR) coloring game designed to enhance creativity and engagement. In this game, you control a virtual eagle that changes its appearance based on the color you create on a physical paper. The eagle then interacts with various animals in a dynamic AR jungle environment. This manual will guide you through the different scenes, controls, and overall gameplay.

2. System Requirements

- **Device: A mobile device (smartphone or tablet) that supports AR Foundation.**
- **Operating System: Android (or iOS, if applicable).**
- **Camera: A functioning rear camera for AR scanning.**
- **Internet: (Optional) For downloading additional assets if needed.**

3. Game Flow Overview

The game consists of multiple scenes with distinct functions:

1. **Main Menu:**
   - **Displays the game title “The Journey of the Eagle” and a “PLAY” button.**
   - **The background features an animated eagle (flying in a circular pattern).**
2. **AR Cam Scene:**
   - **Activates the AR camera and scans for specific images from a pre-defined library.**
   - **When a target image (e.g., representing "eagleRed", "eagleRGB", "eagleBlue", or "eagleNormal") is detected, the game saves the corresponding eagle skin.**
   - **After a short delay, the scene transitions automatically to the Color Confirmation scene.**
3. **Color Confirmation Scene:**
   - **Displays the eagle with its updated appearance (the mesh and materials are replaced according to the scanned image).**
   - **A prompt asks, “Is this the Color you want?” with two buttons: YES and NO.**
   - **Pressing YES continues to the Main Game scene; pressing NO returns you to the AR Cam scene for a re-scan.**
4. **Main Game Scene:**
   - **Contains gameplay elements such as a fixed joystick for movement, an attack button, and a kill count display.**
   - **The eagle (with its updated skin) engages in combat with various target animals (e.g., Sheep, Cow, Chicken, Duck, Goat).**
   - **Each target animal has health, and when its health depletes, it is destroyed and later respawns after a delay.**

4. How to Play

4.1. Main Menu

- **Overview:**
  The Main Menu welcomes you with the game title and a dynamic background featuring the eagle.
- **Action:**
  Tap the PLAY button to start the game. This will load the AR Cam Scene.

4.2. AR Cam Scene (Scanning)

- **Overview:**
  In the AR Cam Scene, the camera view is active. Hold your device so that the target image (printed on paper) is visible in the camera's view.
- **Functionality:**
  - **The app uses AR scanning to detect a specific image from its library.**
  - **When a valid target image is recognized (for example, one that represents a particular eagle skin like "eagleRed"), the app saves that selection.**
  - **On-screen status and debug messages inform you about the scanned image.**
- **Action:**
  Simply point your device at the target image. When you see a confirmation message (e.g., "Saved Eagle: eagleRed"), the system will wait briefly (about 1 second) before automatically transitioning to the next scene.

4.3. Color Confirmation Scene

- **Overview:**
  The Color Confirmation Scene displays the eagle with the new appearance (the mesh and materials have been replaced to match the scanned image).
- **Functionality:**
  - **The screen shows the prompt "Is this the Color you want?"**
  - **Two buttons are available: YES and NO.**
- **Action:**
  - **Press YES if you are happy with the eagle's appearance. This will proceed to the Main Game Scene.**
  - **Press NO if you want to re-scan, which will take you back to the AR Cam Scene.**

4.4. Main Game Scene

- **Overview:**
  In the Main Game Scene, you control the eagle as it interacts with various target animals.
- **Controls:**
  - **Joystick: Use the on-screen fixed joystick to move the eagle.**
  - **Attack Button: Tap the Attack button to initiate an attack. The eagle will automatically target the nearest enemy.**
  - **Menu Button: Use this button to return to the Main Menu.**
- **Gameplay:**
  - **When the eagle attacks a target, the target's health decreases.**

    - **When a target's health reaches zero, it is destroyed, and the ScoreManager updates the kill count.**
    - **The TargetRespawner script schedules the target to respawn after a 20-second delay.**
- **Action:**
  Use the joystick to move and the attack button to defeat targets. Watch the kill count increase as you defeat enemies.

5. Detailed Features & System Modules

5.1. AR Scanning & Model Replacement

- **ARImageColorSceneSwitcher / EagleManager:**
    - **Scans AR images and saves the selected eagle skin (using PlayerPrefs).**
    - **Replaces the default eagle's mesh and material (using the SkinnedMeshRenderer's sharedMesh and sharedMaterials) without changing its Transform.**
    - **This module ensures that the eagle's new appearance (e.g., eagleRed) is consistently used in subsequent scenes.**

5.2. Target Attack & Respawn

- **EagleAttackController:**
    - **Dynamically finds the nearest target (any object with a Target script) and commands the eagle to attack.**
    - **Plays attack animations, sequential audio, and visual effects upon hitting a target.**
- **Target & TargetRespawner:**
    - **Each target animal has a health value and is destroyed when its health reaches zero.**
    - **The TargetRespawner manages the delayed respawn (e.g., 20 seconds) of each target.**

5.3. UI & Scene Navigation

- **ScoreManager:**
    - **Updates the kill count displayed on the Canvas whenever a target is destroyed.**
- **ChangeScene:**
    - **A simple script that transitions between scenes based on user interactions (e.g., from Main Menu to AR Cam, Color Confirmation, and Main Game).**
- **SmoothCameraFollow:**
    - **Provides smooth camera motion in the Main Game Scene, ensuring that the camera follows the eagle seamlessly.**

6. Design Evolution and Key Decisions

6.1. Requirements Gathering and Paper Prototyping

- **Initial Concept:**
  The early idea was to create a simple AR coloring application. During requirements gathering, feedback emphasized the importance of integrating social interaction and dynamic model replacement.

- **Paper Prototypes:**
  Early sketches highlighted the need for clear scene transitions and intuitive AR scanning feedback. Several layout options were considered for the AR Cam and Color Confirmation scenes.

6.2. Heuristic Evaluation and Iterative Refinement

- **AR Scanning & Model Replacement Accuracy:**
  Initial tests showed that users were uncertain whether the scanned image was correctly recognized. We added real-time on-screen status (statusText) and debug messages (debugText) to address this issue.
- **Model Replacement Stability:**
  During heuristic evaluations, users noted that the replaced eagle model sometimes appeared inconsistent. We resolved this by directly replacing the SkinnedMeshRenderer's sharedMesh and sharedMaterials and setting updateWhenOffscreen = true to ensure consistent display even when the model is offscreen.
- **User Interface Adjustments:**
  The layout of the Color Confirmation scene was modified based on user feedback to make the YES/NO choices more prominent. We also improved the overall clarity of debug information.
- **Alternative Considerations:**
  We considered dynamically modifying material color versus replacing the entire mesh. Mesh replacement was chosen because it preserves the full visual detail of the eagle model, including textures and animations.

6.3. Future Enhancements

- **AI-Powered Personalization:**
  Future work will integrate AI-driven personalization to recommend eagle skins based on user behavior.
- **Platform Expansion:**
  The next phase will focus on optimizing performance for a wider range of devices and enhancing multiplayer interactions.
- **UI and Interaction Enhancements:**
  Additional user testing will help refine UI layouts and interactions further.

7. Illustrations and Screenshots

(Include the following screenshots in your final report or document them as needed):

- **Main Menu Screenshot:**
  Show the game title "The Journey of the Eagle" and the PLAY button.
- **AR Cam Scene Screenshot:**
  Display the AR camera view with on-screen statusText and debugText showing scan results.

- **Color Confirmation Scene Screenshot:**
  Show the default eagle with its replaced mesh and the YES/NO buttons for confirmation.
- **Main Game Scene Screenshot:**
  Display the gameplay interface with the joystick, attack button, kill count UI, and a jungle background, along with visual feedback of target interactions.

8. Summary

The final implementation meets all the functional requirements, including accurate AR scanning, dynamic model replacement, target attack with respawn, and smooth scene transitions. The system's design evolved iteratively through requirements gathering, paper prototyping, and heuristic evaluations. Key design decisions—such as choosing mesh replacement for model switching and using a centralized respawn manager—were made based on user feedback. Future enhancements will focus on further personalizing the user experience and expanding system compatibility.